NAME :

UMER

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SUBJECT :

PF

SUBMIT BY :

SIR ASAD

CLASS :

BSCS (E)

TASK 1:-

```cpp
#include<iostream>
using namespace std;
int main()
{

    int num,i,tab;
    cout<<"Enter any number for table:";
    cin>>num;

    cout<<"Table of "<< num <<" is:"<<endl;
    cout<<endl;
    for(i=1;i<=10;i++)
    {
        tab = num * i;
        cout<<num <<" * "<<i<<" = "<<tab<<endl;

    }

    return 0;

}
```

OUTPUT:-

```
Enter any number for table:2
Table of 2 is:

2 * 1 = 2
2 * 2 = 4
2 * 3 = 6
2 * 4 = 8
2 * 5 = 10
2 * 6 = 12
2 * 7 = 14
2 * 8 = 16
2 * 9 = 18
2 * 10 = 20
```

TASK 2:-

```cpp
#include<iostream>
using namespace std;
int main()
{
    int base,power;
    long result = 1;
    cout<<"Enter the base number:";
    cin>>base;
    cout<<"Enter the power:";
    cin>>power;

    for(int i=1;i<=power;i++)
    {
        result = result * base;
    }

    cout<<result;



    return 0;
}
```

OUTPUT:-

```
Enter the base number:2
Enter the power:3
8
--------------------------
```

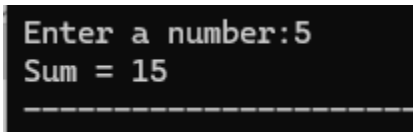
TASK 3:-

```
#include<iostream>
using namespace std;
int main()
{
    int sum,i,num;
    sum = 0;
    cout<<"Enter a number:";
    cin>>num;

    for(i=1;i<=num;i++)
    {
        sum = sum + i;
    }

    cout<<"Sum = "<<sum;
    return 0;
}
```

OUTPUT:-



TASK 4:-

```cpp
#include<iostream>
using namespace std;
int main()
{
    int num,digit;
    int sum = 0;
    cout<<"Enter a number: ";
    cin>>num;

    while(num>0)
    {
        digit = num % 10;
        sum = sum + digit;
        num = num / 10;
    }

    cout<<"Sum of number is: "<<sum;
    return 0;
}
```

OUTPUT:-

```
Enter a number: 789
Sum of number is: 24
--------------------
```

TASK 5:-

```cpp
#include<iostream>
using namespace std;
int main()
{
    int num;
    long factorial = 1;

    cout<<"Enter positive number : ";
    cin>>num;

    if(num<0)
    {
        cout<<"You entered wrong number!";
    }
    else
    {
        for(int i=1;i<=num;i++)

        factorial = factorial *i;

    }
    cout<<num<<" ka factorial hai : "<<factorial;
    return 0;
}
```

OUTPUT :-

```
Enter positive number : 5
5 ka factorial hai : 120
--------------------------------
```

TASK 6:-

```cpp
#include<iostream>
using namespace std;
int main()
{

    cout<<"*"<<endl;
    cout<<"**"<<endl;
    cout<<"***"<<endl;
    cout<<"****"<<endl;
    cout<<"*****"<<endl;
    cout<<"******"<<endl;
    cout<<"*******";

    return 0;
}
```

OUTPUT:-

```
*
**
***
****
*****
******
*******
```

TASK 7:-

```cpp
#include <iostream>
using namespace std;

int main()
{
    int month;

    cout << "Enter month number (1-12): ";
    cin >> month;

    switch(month)
    {
        case 1:
        case 3:
        case 5:
        case 7:
        case 8:
        case 10:
        case 12:
            cout << "31 days" << endl;
            break;

        case 4:
        case 6:
        case 9:
        case 11:
            cout << "30 days" << endl;
            break;

        case 2:
            cout << "28 or 29 days (Leap year)" << endl;
            break;

        default:
            cout << "Invalid month number!" << endl;
    }

    return 0;
}
```

OUTPUT:-

```
Enter month number (1-12): 1
31 days
```

TASK 8:-

```cpp
#include<iostream>
using namespace std;
int main()
{
    char alfabet;
    cout<<"Enter Any Character:";
    cin>>alfabet;
    switch(alfabet)
    {
        case 'a':
            cout<<"You entered vowel";
            break;
        case 'e':
            cout<<"You entered vowel";
            break;
        case 'i':
            cout<<"You entered vowel";
            break;
        case 'o':
            cout<<"You entered vowel";
            break;
        case 'u':
            cout<<"You entered vowel";
            break;
        default:
            cout<<"You entered a consonant";

    }
    return 0;
}
```

OUPUT:-

```
Enter Any Character:A
You entered a consonant
--------------------------
```

OUTPUT OF PROGRAMS:-

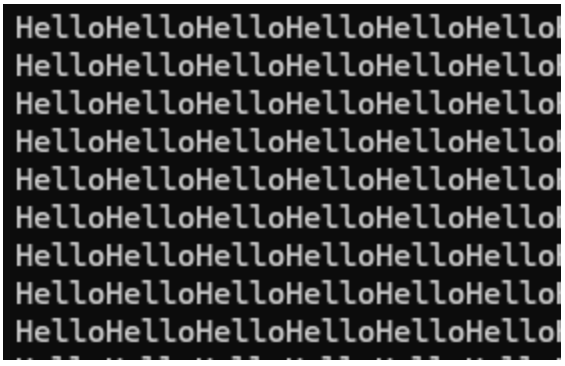
FIRST:-

```
#include<iostream>
using namespace std;

int main()
{
//  Error on loop condition
    for(5;2;2)
    {
        cout<<"Hello";
    }
    return 0;
}
```

OUTPUT:-

INFINITE LOOP



SECOND:-

```cpp
#include<iostream>
using namespace std;

int main()
{
    int i;
//  Error in Condition
    for(i=0;i>9;i+=3)
    {
        cout<<"for ";
    }
    return 0;
}
```

OUTPUT:-

NO OUPUT



THIRD:-

```cpp
#include<iostream>
using namespace std;

int main()
{
//  Ifinite loop . Error Ocuur
    for(;;)
    {
        cout<<"10";
    }
    return 0;
}
```

OUTPUT:-

INFINITE LOOP



FOURTH:-

```
#include<iostream>
using namespace std;

int main()
{
    int fun=5;
    cout<<"C++ for loop";
    int x = 5;
    for(x=0;x<=fun;x++)
    {
        cout<<x;
    }
    return 0;
}
```

OUTPUT:-


C++ for loop012345